DO NOW — RECALL

Lesson 085 · Searching, Sorting and Pathfinding Compar

Name: _____ Class: _____ Date: _____

1. What is the sorted-input rule for binary search?

2. State merge sort's worst-time complexity.

3. State f score in A*.

4. Push A,B,C onto a stack, Pop twice, then Push D. Give bottom-to-top order.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

The input must be sorted.

2. Model answer

O(n log n).

3. Model answer

f=g+h.

4. Model answer

Pop removes C then B; the final stack is [A,D] bottom-to-top.

By the end, you can…

• Start from the problem and input: searching, ordering and weighted paths are different problem classes.

• A fastest method under one set of assumptions may be invalid or less suitable on another dataset; include preprocessing and memory costs.

• Choose using correctness, worst-case growth, memory, stability where relevant, and an independent trace on new values.

Vocabulary — know these terms

suitability — How well method fits input/task.

precondition — Requirement for correct use.

complexity — Growth of time/space.

heuristic — Estimate guiding search.

NEW LEARNING

Read, annotate and explain

Observe

Start from the problem and input: searching, ordering and weighted paths are different problem classes.

Apply

A fastest method under one set of assumptions may be invalid or less suitable on another dataset; include preprocessing and memory costs.

Evaluate

Choose using correctness, worst-case growth, memory, stability where relevant, and an independent trace on new values.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Choose algorithm for 200 sorted pupils by ID with repeated lookups.

Method and result

Binary search O(log n) is valid because identifiers are sorted; if indexed dictionary is available, keyed lookup is another option.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Identify suitable algorithm for (a) unsorted match (b) sorted match (c) non-negative weighted shortest path. [6]

Question 2

Compare bubble, insertion, merge and quick sort worst-case time and typical auxiliary memory. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

A college searches a million unsorted attendance records once and later queries sorted year-group reports repeatedly. Evaluate algorithms/data processing choices. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

All standard algorithms can be swapped freely without checking their input assumptions.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.